Day 1

Number Systems - Assignment

Q1) Complete the table with equivalents in Binary, Octal and Hexa for: 143.75 (decimal), 11011.101 (binary), 453 (octal), 1F.C(hex).

i) 143.75 (decimal) to binary

$143.75_{(10)} \Rightarrow (?)_{(2)}$

$$\begin{array}{r|ll}
2 & 143 & \\
2 & 71 & -1 \\
2 & 35 & -1 \\
2 & 17 & -1 \\
2 & 8 & -1 \\
2 & 4 & -0 \\
2 & 2 & -0 \\
 & 1 & -0
\end{array}$$

$0.75 \times 2 = 1.5 - 1$

$0.5 \times 2 = 1.0 - 1$

$0 \times 2 = 0 - 0$

$\therefore 143.75_{(10)} = 10001111.1100_{(2)}$

$143.75_{(10)} \Rightarrow (?)_8$

$$\begin{array}{r|ll}
8 & 143 & \\
8 & 17 & -7 \\
8 & 2 & -1 \\
 & 0 & -2
\end{array}$$

$0.75 \times 8 = 6 - 6$

$\therefore 143.75_{(10)} = 217.6_{(8)}$

$143.75_{(10)} \rightarrow (?)_{(16)}$

$$\begin{array}{r|l} 16 & 143 \\ 16 & 8 - 15(F) \\ & 0 - 8 \end{array}$$

$0.75 \times 16 = 12 \quad - \ C$

$\therefore 143.75_{(10)} = \underline{\underline{8F.C_{(16)}}}$

ii) $11011.101_{(2)} \rightarrow (?)_{(10)}$

$\Rightarrow 1\times 2^4 + 1\times 2^3 + 1\times 2^1 + 1\times 2^0 \qquad 1\times 2^{-1} + 1\times 2^{-3}$

$16 + 8 + 2 + 1 \qquad 0.5 + 0.125 = 0.625$

$\therefore 11011.101_{(2)} = \underline{\underline{27.625_{(10)}}}$

$\underbrace{011}\underbrace{011}.\underbrace{101}_{(2)} \rightarrow (?)_8$

$\therefore 11011.101_{(2)} = \underline{\underline{33.5_{(8)}}}$

$\underbrace{0001}\underbrace{1011}.\underbrace{1010}_{(2)} = \underline{\underline{1}}B.A_{(16)}$

$\therefore 11011.101_{(2)} = \underline{\underline{1B.A_{(16)}}}$

iii) $453_{(8)} \rightarrow (?)_{(10)}$

$$4 \times 8^2 + 5 \times 8^1 + 3 \times 8^0$$

$$256 + 40 + 3$$

$$453_{(8)} = \underline{\underline{299_{(10)}}}$$

$$453_{(8)} \rightarrow (?)_{(2)}$$

$$453_{(8)} = \underline{\underline{100101011_{(2)}}}$$

$$453_{(8)} \rightarrow (?)_{(16)}$$

$$453_{(8)} = \cancel{01000101001} \underbrace{0001}\ \underbrace{0010}\ \underbrace{1011}$$

$$453_{(8)} = \underline{\underline{12B_{(16)}}}$$

iv) $\overset{1}{1}\overset{0}{F}.\overset{-1}{C}_{(16)} \rightarrow (?)_{(10)}$

$$1 \times 16^1 + 15 \times 16^0 + 12 \times 16^{-1}$$

$$16 + 15$$

$$1F.C_{(16)} = \underline{\underline{31.75_{(10)}}}$$

$1F.C_{(16)} \rightarrow (?.)_{(2)}$

$1F.C_{(16)} = 00011111.1100_{(2)}$

$1F.C_{(16)} \rightarrow (?.)_{8}$

~~$1F.C_{(16)} = 00011111.1100$~~

$1F.C_{(16)} = \underbrace{000}_{0}\underbrace{011}_{3}\underbrace{111}_{7}.\underbrace{110}_{6}\underbrace{000}$

$\therefore 1F.C_{(16)} = 37.60_{(8)}$

Q2) Find equivalents:

a) $324_{(5)} = (?.)_{(10)}$

$3\times5^{2} + 2\times5^{1} + 4\times5^{0}$

$75 + 10 + 4$

$\therefore 324_{(5)} = 89_{(10)}$

b) $32.23_{(4)} = (?.)_{(10)}$

$3\times4^{1} + 2\times4^{0} + 2\times4^{-1} + 3\times4^{-2}$

$= 12 + 2 + 0.5 + 0.1875$

$\therefore 32.23_{(4)} = 14.6875_{(10)}$

c) $231_{(10)} = (?.)_{(7)}$

$$\begin{array}{r|l} 7 & 231 \\ \hline 7 & 33 - 0 \\ \hline & 4 - 5 \end{array}$$

$\therefore 231_{(10)} = 450_{(7)}$

d) $189_{(10)} = (?.)_6$

$$\begin{array}{r|l} 6 & 189 \\ \hline 6 & 31 - 3 \\ \hline & 5 - 1 \end{array}$$

$\therefore 189_{(10)} = 513_{(6)}$

Q3) Find the radix r:

a) $(312/20)_r = (13.1)_r$

Taking LHS,

$(312)_r = 3r^2 + r + 2$

$(20)_r = 2r$

$$\Rightarrow \frac{3r^2+r+2}{2r}$$

Taking RHS,

$$(13.1)_r = 1 \times r + 3 + \frac{1}{r}$$

$$= r + 3 + \frac{1}{r}$$

Equating both sides,

$$\frac{3r^2+r+2}{2r} = r+3+\frac{1}{r}$$

$$3r^2+r+2 = 2r^2+6r+2$$

$$r^2-5r=0$$

$$r(r-5)=0$$

$$r=0 \quad \text{or} \quad r=5$$

$$\therefore \; r=5$$

b) $264_r + 136_r = 433_r$

$$\Rightarrow \quad 264_r = 2r^2+6r+4$$

$$136_r = r^2+3r+6$$

$$433_r = 4r^2+3r+3$$

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$(2r^2+6r+4) + (r^2+3r+6) = 4r^2+3r+3$

$3r^2+9r+10 = 4r^2+3r+3$

$r^2-6r-7=0$

$(r-7)(r+1)=0$

$r=7$ or $r=-1$

$\underline{\underline{r=7}}$ ($\because$ radix cannot be negative).

c) $221505_8 = 12345_r$

$\Rightarrow$

$221505_8 = 2\times8^5 + 2\times8^4 + 1\times8^3 + 5\times8^2 + 5\times8^0$

$= 2(32768) + 2(4096) + 512 + 5(64) + 5$

$= 65536 + 8192 + 512 + 320 + 5$

$12345_r = \underline{74565}_{10}$

$12345_r = r^4 + 2r^3 + 3r^2 + 4r + 5$

$74565 = r^4 + 2r^3 + 3r^2 + 4r + 5$

$\because$ largest digit is 5, $r > 5$

Let $r = 16$:

$16^4 + 2(16^3) + 3(16^2) + 4(16) + 5$

$= 65536 + 8192 + 768 + 64 + 5$

$= \underline{74565}$

$\therefore$ $\underline{r=16}$

Q4) Addition:

a) $1101_2 + 1101111_2$

$\Rightarrow$

$$
\begin{array}{r}
00001101 \\
+\ 1101111 \\
\hline
1111100 \\
\end{array}
$$

b) $77652_8 + 76543_8$

$\Rightarrow$

$$
\begin{array}{r}
77652 \\
76543 \\
\hline
176415 \\
\end{array}
$$

$9 = \underset{\text{carry}}{1} \times 8 + \underset{\text{sum}}{1}$

$12 = 1 \times 8 + 4$

$14 = 1 \times 8 + 6$

$15 = 1 \times 8 + 7$

$\therefore 77652_8 + 76543_8 = 176415_8$

c) $578A_{16} + ABDE_{16}$

$\Rightarrow$

$$
\begin{array}{r}
578A \\
ABDE \\
\hline
10368 \\
\end{array}
$$

$A + E = 10 + 14 = 24$

$24 = 1 \times 16 + 8$

$22 = 1 \times 16 + 6$

$19 = 1 \times 16 + 3$

$16 = 1 \times 16$

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$\therefore 578A_{16} + ABDE_{16} = 10368_{16}$

d) $234_6 + 315_6$

$\Rightarrow$

$$\begin{array}{r} 234 \\ 315 \\ \hline 553 \end{array}$$

$9 = 1 \times 6 + 3$

$\therefore 234_6 + 315_6 = 553_6$

Q5) Binary $\rightarrow$ Gray code:

$A\bar{B} + \bar{A}B$

$1 \cdot 1 + 0 \cdot 0$

1

a) 10101_2

$\Rightarrow$ Binary : 1 0 1 0 1

Gray : 1 1 1 1 1

$1 \oplus 0 = 1$

$0 \oplus 1 = 1$

$1 \oplus 0 = 1$

$0 \oplus 1 = 1$

$\therefore 10101_2 = 11111_{Gray}$

b) 01110_2

$\Rightarrow$ Binary: 0 1 1 1 0

Gray: 1 0 0 1 0 $\leftarrow$

$A\bar{B}+\bar{A}B$

$1.0+0.1$

$0+0$

$1 \oplus 0 = 1$

$1 \oplus 1 = 0$

$1 \oplus 1 = 0$

$0 \oplus 1 = 1$

$\therefore 01110_2 = 01001_{Gray}$

Q6) Gray code $\rightarrow$ Binary:

a) 01101 (gray)

$\Rightarrow$ Gray: 0 1 1 0 1

Binary: 0 1 0 0 1 $\rightarrow$

~~$1 \oplus 0 = 1$~~	$0 \oplus 1 = 1$
~~$1 \oplus 0 = 1$~~	$1 \oplus 1 = 0$
~~$1 \oplus 1 = 0$~~	$0 \oplus 0 = 0$
~~$0 \oplus 1 = 1$~~	$0 \oplus 1 = 1$

$\therefore 01101_{Gray} = 01001_2$

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b) 10101 (gray)

=>

Gray : 1 0 1 0 1

Binary : 1 1 0 0 1

$1 \oplus 0 = 1$

$1 \oplus 1 = 0$

$0 \oplus 0 = 0$

$0 \oplus 1 = 1$

$\therefore 10101_{gray} = \underline{\underline{11001_2}}$